



How Pedestrians React to Imminent Vehicle Threats: A Virtual Reality Study in Safety-Critical Scenarios

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Abstract—The uncertainty of pedestrians’ behaviors under safety-critical scenarios poses a challenge for highly automated vehicles (HAVs) decision-making. When faced with imminent vehicles, pedestrians need to balance their road crossing desire with the demands for safety, resulting in either normal walking or avoidance behaviors. Therefore, it is important to analysis such behaviors so that HAVs can better understand pedestrians and develop safe interaction algorithms or protective strategies. To study these behaviors, this paper designed an immersive virtual reality (VR) experiment under controlled conditions where participants encountered traffic scenarios with different spatiotemporal pressures. The experiment has revealed three distinct behavior modes under increasing spatiotemporal pressures: a no-risk mode driven solely by the desire of crossing road, a game mode balancing safety and crossing desire, and a life-saving mode prioritizing collision avoidance. Based on these behavior modes, we proposed a regression model to predict pedestrian decisions. The model achieved average precision of 0.87 and 0.82 for predicting whether and how pedestrian avoid. Finally, real pedestrian-vehicle conflict data are used to validate the effectiveness of the experimental results. This investigation clarifies pedestrians’ behavior modes in various urgent scenarios and presents a decision model based on pedestrians’ behavior modes, detailing their inner interaction logic with hazardous vehicles in safety-critical scenarios. We expect that the observed behavior modes and decision model can help develop safety algorithms of HAVs to make safe interactions with pedestrians on roads.

Index Terms—Highly automated vehicle, Pedestrian safety, Pedestrian behavior, Safety-critical scenario, Virtual reality experiment

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I. INTRODUCTION

PEDESTRIANS are vulnerable road users and face high injury risk when interacting with vehicles [1]. Ensuring pedestrian safety remains a critical prerequisite for implementation of HAVs [2]. As a research hotspot in the field of intelligent traffic, understanding and modelling pedestrian behaviors is paramount for enhancing the interaction safety between HAVs and pedestrians [3-6].

Specific to safety-critical scenarios, pedestrians invariably face a dilemma of balancing their road crossing desire and safety assurance, leading them to make decisions that either maintain or change their movement [7]. Generally, they have to make above decisions when vehicles are approaching. These decisions without sufficient thought may lead to highly unpredictable behaviors [8]. Such uncertain behaviors complicate HAVs’ understanding of pedestrians, increasing the collision risk. Particularly, as simple driver assistance tools evolve towards self-driving [9], there is a growing need for a deep comprehension of how pedestrians make decisions. This demand is not satisfied just by the current data-driven approaches. Instead, it requires an understanding of pedestrian behaviors from cognitive and empirical sciences [10]. Because algorithms that incorporating pedestrian inner behavioral logic enable HAVs to interact with pedestrians in a human-like way. This, leads to the research context of this study, which is to understand pedestrian behaviors and model their decisions in safety-critical scenarios. Below, we review the relevant phenomenon, theories and results to understand pedestrian behaviors in safety-critical scenarios.

A. Pedestrian behaviors in safety-critical scenarios

Pedestrian behaviors in safety-critical scenarios arise from unexpected vehicle appearances due to pedestrians’ visual obstructions [11] or attention shifts [12]. In such situations, pedestrians must assess the available information to decide whether to cross as normal or actively avoid the vehicle [13]. Many studies have focused on this phenomenon by collecting pedestrian behaviors through simulated accident experiments [14], [15] or VR experiments [16]. Their findings have revealed that four common behaviors occur in such scenarios: normal walking (i.e., pedestrians show no reaction or perceive no need to avoid), stopping (i.e., pedestrians freeze from shock, or they have not yet entered the hazardous zone), forward accelerating (i.e., pedestrians accelerate to escape the

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hazardous zone), backward avoidance (i.e., pedestrians retreat by changing direction to exit the hazardous zone). These behaviors directly affect the pedestrian’s movement, influencing the possibility of a collision. However, due to the limited interaction data in safety-critical scenarios, current researches on such behaviors are limited to case analyses.

B. Datasets for pedestrian behaviors

To understand inner logic of pedestrian behaviors, a large, high-quality dataset is needed, particularly when they include human-annotated “ground truth information”. However, in-depth accident study datasets primarily document collision causes and pedestrian injuries, with insufficient data on pedestrian behaviors before collision [17]. Data from driving recorder, drones, and traffic cameras captures extensive interaction behaviors between pedestrians and vehicles [18-20]. But their limited coverage of hazardous pedestrian-vehicle interactions hinders the understanding of pedestrian behaviors in safety-critical scenarios. VR-based pedestrian simulators are employed to gather detailed data on pedestrian-vehicle interactions, focusing on pedestrians’ behaviors in specific traffic scenarios [21-23]. For example, pedestrian crossing behaviors in single pedestrian-vehicle scenarios were investigated through external human machine interfaces [24]. More significantly, virtual hazardous vehicles were used to stimulate pedestrians, capturing their behaviors during interactions [16]. The above research indicates that pedestrian simulators may be an effective way to solve the lack of high-quality data in safety-critical scenarios.

C. Cognitive theory for human behaviors under stimuli

In order to understand pedestrian inner behavioral logic. Studies often simplify hazardous pedestrian-vehicle interaction to a decision task under temporal or spatial stimuli. Specifically, human can gather enough information under low time pressure to make near-optimal decisions [25], as exemplified by pedestrians choosing normal walking in safe scenarios. However, under high time pressure, limited information acquisition often leads to higher error rates in decisions [26], for example, pedestrians often exhibit increased misjudgment in safety-critical scenarios, which leads to collisions. More interestingly, human decision uncertainty demonstrates a nonlinear relationship with the urgency of a stimuli. Uncertainty exhibits high levels in both extremely urgent and low-urgent scenarios. Conversely, it tends to be relatively low under intermediate scenarios [8]. These conclusions are drawn from specific experimental paradigms in cognitive science. However, due to substantial differences between cognitive paradigms and real traffic conditions, these theories have not been directly applied to understand pedestrian behavior logic in real scenarios.

D. Computation models for pedestrian decisions

Pedestrian decisions determine their subsequence behaviors.

Many studies seek to model how pedestrian make decisions and interact with vehicles through traffic data. Among them, the probability-driven approach offers a probabilistic description of pedestrian decisions. For instance, probabilistic graphical network and navigation function were used to model pedestrian crossing decision at signalized intersections [27], [28]. The theory-driven approach describes pedestrian decisions based on mathematical and physical theory. For example, the discrete models simulate pedestrian decisions by discretizing traffic space and establishing movement rules [29]. Continuous models, such as social force model, use differential equations to represent forces that influence pedestrian movements and decisions [30]. Furthermore, game theory provides a robust framework for modeling dynamic pedestrian decisions. For example, discrete game framework involving decisions to yield or maintain speed was used to model pedestrian decisions at unsigned intersections [21]. However, the probability-driven and theory-driven approaches primarily focus on pedestrian crossing scenarios.

In the safety-critical scenarios, the dynamic nature of pedestrian decisions and the urgency of time pressure necessitate further research into how pedestrian make decisions when interact with vehicles [10]. The data-driven approach utilizes interaction data between pedestrians and vehicles, using heuristic algorithms to model pedestrian decisions. For instance, imitation learning and reinforcement learning utilizes pedestrian trajectory and speed data to simulate complex pedestrian decisions [31], [32]. Such methods depend significantly on training datasets, and their interpretability and generalization require further study. In summary, while traffic data can support to employs mathematical models to analyze pedestrian decisions, its effectiveness relies heavily on the available dataset. Therefore, to develop a pedestrian decision model with inner behavioral logic under safety-critical scenarios, it is necessary to combine computation models with cognitive theory analysis.

E. Research gap

The uncertain behaviors of pedestrians in safety-critical scenarios significantly impacts the safe interaction between HAVs and pedestrians. Consequently, it is essential to comprehend and model these behaviors to further enhance HAVs safety algorithm. However, due to the above discussion, this phenomenon has not yet been focused, which can be summarized as follows:

- (i) There is a lack of high-quality pedestrian behavior datasets under safety-critical scenarios.
- (ii) There is a lack of cognitive theory-based analysis of pedestrians’ behaviors in safety-critical scenarios, and there is a need for a more thorough explanation of the uncertainty in their behaviors.
- (iii) There is currently lack of a pedestrian decision computational model that incorporates cognitive theory-based analysis to describe pedestrians’ behaviors in safety-critical scenarios.

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F. Contribution of this study

Consequently, the present study was aimed to analyze pedestrian behavior modes from the perspective of cognitive theory and to develop a preliminary pedestrian decision model in safety-critical scenarios (Fig. 1). The problem was decomposed into three parts:

(i) Establish a pedestrian-in-loop virtual interactive platform, design typical safety-critical scenarios, and gather data on pedestrian-vehicle interactions from an experimental program using human subjects.

(ii) Analyze pedestrian behaviors under various imminent collision scenarios and different time urgency levels to investigate the inner behavioral logic and classify pedestrian behavior modes.

(iii) Develop a decision model to describe pedestrian behaviors in different urgent safety-critical scenarios.

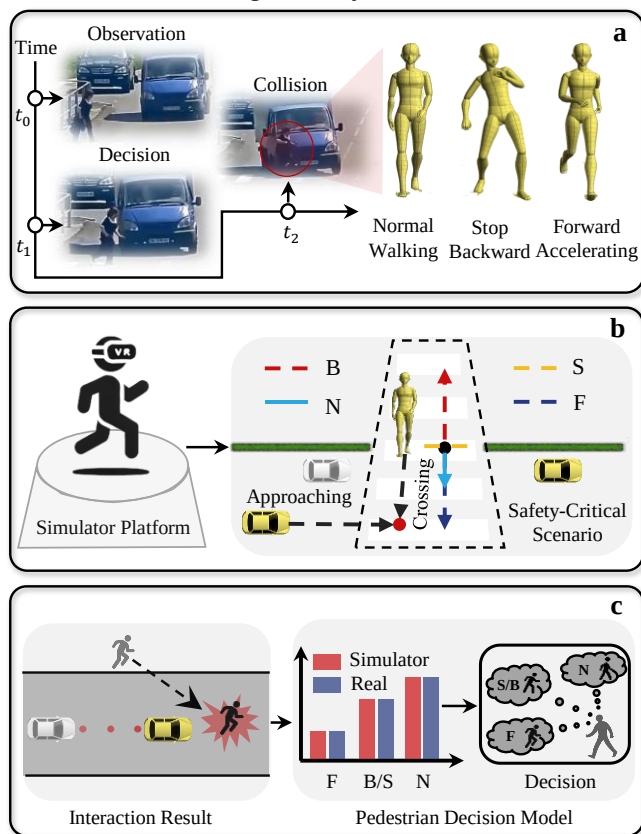


Fig. 1. Framework of pedestrian behaviors analysis and modelling under safety-critical traffic scenarios. (a) Pedestrian decision process and behaviors in real-world scenarios. (b) Virtual reality experiment with pedestrian-in-loop. (c) Pedestrian behavior modes and decision modeling.

The remainder of the article is organized as follows. Section II details the establishment and result of pedestrian-in-loop virtual reality experiment. Then, Section III analyzes pedestrian behaviors through statistical and cognitive methods, and classifies the behavior modes of pedestrians in different scenarios. Section IV introduces a regression model and evaluation metrics to describe pedestrian decisions under different behavior modes. Section V is the discussion, which are then concluded in Section VI.

II. PEDESTRIAN-IN-LOOP VIRTUAL REALITY EXPERIMENT

A. Design and set-up of experimental system

1). Set-up of experiment system.

This study used virtual reality experiments to examine pedestrian's interaction and avoidance behaviors in response to approaching vehicles (Fig. 2a, b). Well-defined scenarios, including a pedestrian (i.e., the subject himself), hazardous vehicle, and traffic flow (Carla 0.9.15) were provided to a human subject. The subject, who wore a VR helmet (HTC Vive Pro 1.0, resolution: 1440×1660 , rendering rate: 90 Hz), entered the virtual environment as a pedestrian. A walking algorithm on the motion platform (KAT WALK 3DT) captured subject's behaviors, such as normal walking (N), stopping (S), forward accelerating (F), and backward avoidance (B), which were then replicated to the pedestrian in the virtual environment through Unreal Engine. Meanwhile, a portable eye-tracking device (Eye Tracker: 7invensus aSee VR, sampling frequency: 120 Hz, sampling delay: < 5 ms, accuracy: $< 0.5^\circ$) was incorporated into the VR helmet to record subject's gaze points as the subject observed traffic scenarios. Additionally, a control system module was designed to regulate traffic, managing the appearance of hazardous vehicles. Simultaneously, a data acquisition module recorded information on pedestrians, vehicles, and environmental factors, such as time, velocity, and position (sampling frequency: 12.5 Hz).

2). Scenario design.

According to typical scenarios of pedestrian-vehicle collisions, most accidents occur at crosswalks [33] and mid-block locations [34]. The causes of such accidents can generally be attributed to two factors: (i) pedestrians violate traffic regulations by running red lights or jaywalking without properly checking for approaching vehicles [35]; (ii) pedestrians adhere to traffic rules, but drivers fail to yield at crosswalks due to obstructed visibility or traffic violations [11]. These two factors mainly occur when pedestrians or drivers notice each other too late, leaving insufficient time to avoid collision. Therefore, we designed two types of safety-critical scenarios: mid-block locations and crosswalks (Fig. 2c). In-depth accident data indicate that the impact velocity between pedestrians and vehicles in urban areas generally falls within the range of 20 to 50 km/h [13]. To obtain pedestrian behaviors under different time pressures, the control system positions a hazardous vehicle 18 m away from the pedestrian, moving at speeds of 15, 25, 35 and 45 km/h, respectively. Thus, the corresponding time periods were at levels of 4.8, 2.9, 2.1 and 1.6 s, respectively. It is important to note that while the experiment was designed with four time-to-collision (TTC) levels and fixed approaching positions, the appearance time of the hazardous vehicle is determined randomly based on the pedestrian's position. Specifically, as the width of the hazardous vehicle is 2 m, we define a near interaction zone within 1.5 m of the vehicle's centerline in the crossing

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direction and a relatively distant interaction zone between 1.5 m and 3 m. When the subject reaches a randomly assigned position within either of the above zones during the crossing task, the hazardous vehicle generation program will be activated. Moreover, an obstacle vehicle is also included to block the subject's view, making the scenarios more realistic and hazardous. Finally, to prevent subject's fatigue and learning effects, the direction of the approaching vehicle and the traffic flow were designed to be randomized. The experiment includes 64 distinct scenarios, encompassing two scenario types, four vehicle velocities, two interaction zones, two vehicle directions, and the presence or absence of obstacle vehicles (Table AI).

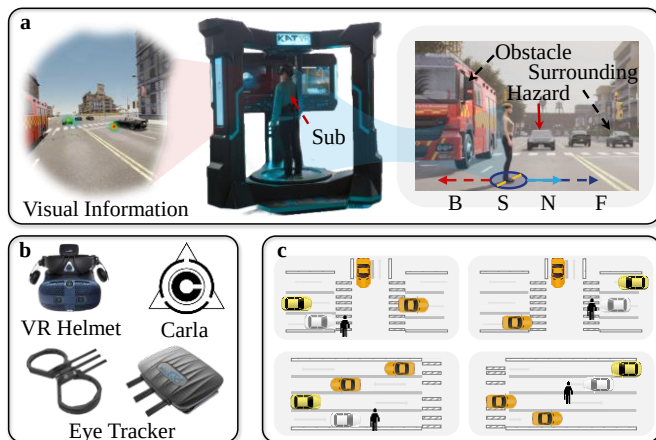


Fig. 2. Experimental design. (a) Virtual reality experiment with pedestrian-in-loop under safety-critical scenarios. (b) Hardware equipment and simulation software. (c) Safety-critical traffic scenarios.

3). Task and Rewards.

A subject entered the virtual scenarios as pedestrian and was instructed to cross the road while ensuring safety. We implemented a reward and punishment mechanism to discourage subject from simply remaining stationary and waiting for vehicles to pass. Specifically, the primary task assigned to the subject was crossing the road. If the subject could walk normally and safely, he would maximize his reward by fulfilling the road crossing. Therefore, successfully crossing the road without a collision while maintaining a normal walking pace resulted in a reward of 100 points for each scenario. On the other hand, pedestrian-vehicle collisions pose severe risks, often resulting in serious injury or even death for pedestrians. To address this concern, when the subject misjudged the danger of the scenario and collided with a hazardous vehicle during the task, a penalty of 100 points was deducted to punish the subject's reckless behavior. In some scenarios, subjects could actively adjust their behavior (i.e., stopping, backward, or forward accelerating) if they perceived that the approaching vehicle was unsafe. This preemptive action aimed to avoid hazardous vehicles and then continued to cross. Consequently, when a subject successfully avoided a vehicle, he was awarded 50 points for the extra

effort to prevent the collision. Conversely, if a collision occurred during the avoidance attempt, 100 points were deducted. This penalty reflected the consequences of wrong decisions, hesitating, or reacting too late, ultimately resulting in a collision. Finally, to ensure high-quality performance, the compensation structure for a subject was directly linked to his total score of all the 64 trials that the subject went through. Each subject who completed all the trials received a base compensation of 320 CNY (Chinese currency). Furthermore, the subjects were eligible for a bonus ranging from 0 to 80 CNY, determined by their ranking relative to all other subjects.

B. Test subject recruitment and test procedure

1). Subject recruitment and ethics statement

Test subjects were recruited through the Shuimu Gang (an online volunteer recruitment platform on campus) from Tsinghua University (population about 100,000). The inclusion criteria specified normal vision (or corrected-to-normal), hearing and walking ability without disabilities or heart disease. Only male individuals between 18 and 30 years old were recruited. A total of 30 subjects (age range 23.5 ± 3.1 years; BMI range 23.0 ± 2.4) were tested in this experimental program and data analysis. The subjects were instructed to crossing the road without collisions while maximizing rewards in each trial. They may withdraw from the experiment if they experience any discomfort. The ethics of this experimental program was reviewed and approved by the Institutional Review Board (IRB) of Tsinghua University, and all the experiments complied with relevant guidelines.

2). Test procedure

The test procedure and equipment (walking platform, eye tracker, VR helmet, control system and data acquisition modules) were operated by the project team members. For each subject, the test process included three phases.

Phase 1. Four scenarios were designed to help the subject familiarize with the virtual environment. A subject, acting as a pedestrian, was given 20 minutes to freely interact with the traffic flow observed in the VR helmet and to adapt to the walking platform ($N = 1.4$ m/s, $F = 2$ m/s, $B = -1.4$ m/s, and $S = 0$ m/s). After the familiarization stage, the subject was asked to stand still and wait for the initialization of the formal experimental scenario.

Phase 2. To prevent subject's fatigue and maintain experimental quality, the 64 scenarios were divided into two sessions, with a subject attending each session on separate days. After initializing the scenario, the subject was positioned at roadside. The subject needed to carefully observe the traffic environment and select an appropriate moment to cross the road. During this process, vehicles in the traffic flow randomly appeared and approached the subject (acting as a pedestrian). Upon noticing a hazardous vehicle, the subject must make decision based on his relative position to the approaching

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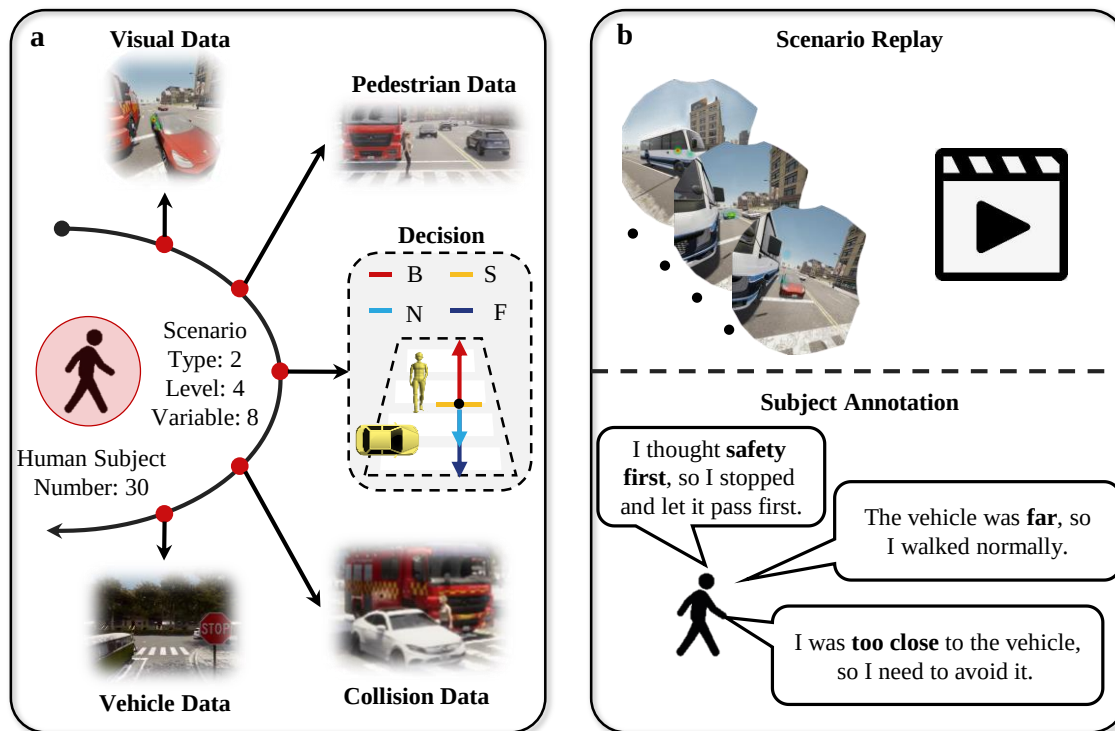


Fig. 3. Dataset of pedestrian-vehicle interactions under safety-critical scenarios. (a) Physical information of pedestrians and vehicles. (b) Human semantic annotation.

vehicle, his estimation of the speed, and the balance of reward and punishment. The experiment concluded once the subject reached the opposite side of the road or collided with the hazardous vehicle. In the process of the experiment, the data acquisition module collected the data, including the subject's visual, motion, and behavior information, as well as details of the hazardous vehicles and the other vehicles in the traffic flow.

Phase 3. A 3-minute break was scheduled between each scenario. During this interval, the test operator identified key interaction moments based on the pedestrian's visual data from the previous scenario. These moments included the appearance of a hazardous vehicle and any behavior changes in the pedestrian's movement. The subject then watched video replays of these critical moments and explained his feelings and behaviors. These explanations will be used as semantic annotations in the study.

C. Data composition

The dataset comprises detailed information of 1920 cases of pedestrian-vehicle interactions in safety-critical scenarios, acquired from the experiments of the 30 human subjects. An eye-tracking system was used for each case to capture the subject's gaze points (Fig. 3a, Visual Data). The pedestrian data and vehicle data acquisition module captured the position, velocity, acceleration, collision information of the pedestrian, the hazardous vehicle, and the surrounding vehicles at each sampling point (Fig. 3a). Furthermore, the control system produced a simulation file after each interaction scenario. This

file enables the test operator to replay the experiment using the Carla simulation software (Fig. 3b, Scenario Replay). Finally, while watching the critical moments in the replay, the subject provided semantic annotations of his feelings and analysis, which were recorded using a voice recorder (Fig. 3b, Subject Annotation).

D. Validity of experiment and dataset

To check the validity of the experimental results, we compared the proportion of pedestrians' decisions with the dataset of 156 real world safety-critical scenarios (i.e., N, F, and no forward avoidance (NF)). B and S were classified as NF). These scenarios, drawn from the DiDi Autonomous Driving Dataset, involve experienced safety drivers in autonomous vehicles. We classify scenarios where the safety driver intervenes as safety-critical, as they have surpassed the driver's risk threshold. In the 156 real-world scenarios, pedestrian decisions were categorized into three distinct types: N (36.8%), F (12.8%), and NF (50.4%). In this experiment, the proportions of subjects' decisions were N (48.3%), F (12.3%), and NF (39.4%). In real and virtual safety-critical scenarios, N and NF are prevalent behaviors. The experimental results indicate that N is slightly higher in virtual scenarios than in real ones, whereas NF is marginally lower (Fig. 4a).

In the experiment, the collision rates for four decisions were N (33.9%), B (25.6%), S (24.4%), and F (17.8%). This suggests that the urgency level in the experiment is appropriate, as it encourages subjects to make rational

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decisions without causing collisions. Additionally, as the velocity of the hazardous vehicle increased from 15 to 45 km/h, the collision rate rose from 1.7% to 67.6% (Fig. 4b). It reveals that, in the experiment, vehicle velocity below 15 km/h can be considered as no hazard, as pedestrian collision rates approach 0. Conversely, velocities exceeding 45 km/h are deemed less meaningful in these virtual scenarios, as pedestrians cannot react timely.

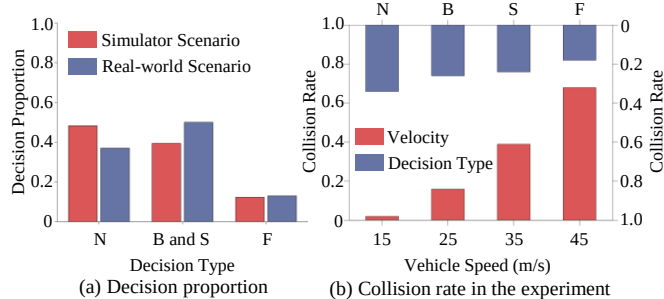


Fig. 4. Validity of the experiment and dataset. (a) Decision proportion between the experiment and the real-world scenarios. (b) Collision rate in the experiment.

III. PEDESTRIAN BEHAVIORS MODES ANALYSIS

Analyzing pedestrian behavior modes under different urgency levels is imperative to understand inner behavioral logic of pedestrians, which involves cognitive aspects. The dataset generated in the experiment provides a comprehensive view of pedestrian decisions in those scenarios. By examining the positions and the conditions at which those decisions are made, we can understand the underlying behavioral logic. Using statistical methods, this section classifies pedestrians' decisions across both temporal and spatial dimensions to uncover the reasons of their behaviors. For consistent comparison, the front center of the hazardous vehicle was taken as the origin, with the x-axis and horizontal direction aligned with the vehicle's approaching direction, and the y-axis and perpendicular direction aligned with the pedestrian's crossing direction (Fig. 1b).

A. Collision rates of pedestrians in different radical styles

1). Define radical styles of pedestrians

The pedestrian behavior modes, influenced by scenarios' urgency levels, dictate the pedestrian decisions. To analyze the subjects' behavior modes, this section introduces two definitions, the "reckless pedestrian" and the "rational pedestrian". The reckless pedestrians lack hesitation, deliberation, or avoidance when facing a hazardous vehicle. They maintain a constant walking velocity of 1.4 m/s and result in the highest collision rate (i.e., the ratio of collisions to interactions within a TTC interval). Such pedestrians are also the target for pedestrian safety protection systems, such as autonomous emergency braking, collision warning system and structural protection measures on vehicles. In contrast, the rational pedestrians refer to those who make a theoretical decision that embodies the ability to make optimal decisions

when crossing roads, maximizing efficiency while ensuring personal safety. This may be hypothetical pedestrian and is a baseline in our experiments, which results in a minimal collision rate. Such pedestrians can be described as:

$$\text{Rational Decision} = \begin{cases} \text{N, if safe} \\ \text{S, if unsafe } y_{\text{rational}} > 1 \\ \text{F, if unsafe } y_{\text{rational}} < 0 \\ \text{B, if unsafe } 0 < y_{\text{rational}} < 1 \end{cases} \quad (1)$$

where y_{rational} represents the perpendicular distance between the rational pedestrian and the hazardous vehicle's centerline. The safety criterion for pedestrians is the expected safe distance, as detailed in Section IV.

As a result of the above definitions, at each decision point of the subjects in the experiment, the decisions and subsequent collision rates were recalculated for reckless and rational pedestrians. This approach determined the upper and lower bounds of collision rates during interactions between pedestrians and hazardous vehicles. To achieve this, based on the TTCs between pedestrians and vehicles, we calculated collision rates for the subjects, reckless pedestrians and rational pedestrians at 0.3 s intervals. These collision rates were then fitted using exponential or polynomial functions.

2). Collision rate analysis for different radical pedestrians

The collision rates of the three radical styles of pedestrians can be described as:

$$\begin{aligned} C_{\text{Rational}} &= 0.8e^{-3.25t}, (R^2 = 0.96) \\ C_{\text{Reckless}} &= 0.95e^{-0.69t}, (R^2 = 0.79) \\ C_{\text{Real}} &= \begin{cases} 1.27t^2 - 1.87t + 0.80, & 0 < t < 1.38, (R^2 = 0.86) \\ 7.16e^{-1.74t}, & 1.38 < t < 4.5, (R^2 = 0.97) \end{cases} \quad (2) \end{aligned}$$

where C_{Rational} , C_{Reckless} , and C_{Real} represent the fitted collision rates for rational pedestrians, reckless pedestrians, and subjects, respectively, and t is the TTC.

The collision rates for both rational and reckless pedestrians consistently decreased as the TTC increased, a trend attributable to their adherence to certain behavior modes. However, the collision rate observed among the subjects displayed two notable shifts. These characteristics divide the decision timeline of pedestrians into four types (Fig. 5). For type 1, TTC is between 0 and t_1 , it is the most significant type because pedestrian decisions during this period directly influence collision rate. Notably, within 0.73 s, the collision rates of the subjects exhibit a similar trend but are marginally higher than those of rational pedestrians. It implies that the subjects' decisions are close to the rational pedestrians. Between 0.73 s and t_1 , the subjects experience a rapid increase in collision rate, close to that of reckless pedestrians. This trend suggests that the subjects begin to try N decisions as TTC increases. Notably, the subjects' avoidance decisions result in a lower collision rate than reckless pedestrians. For type 2, when TTC is between t_1 and t_2 , the collision rates of the subjects twice intersect with those of reckless pedestrians, and overall, the subjects remain generally higher collision

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rates. Although pedestrians can afford more time to weigh their crossing desire against safety, unlike reckless pedestrians, the test subjects often lack decisiveness. This hesitation, especially when wrongly opting for N, leads to higher collision rate. For type 3, when TTC is between t_2 and t_3 , the collision rates of the subjects are relatively low. It reveals that an increased TTC reduces the urgency of the scenario, affording the subjects more time to deliberate decisions. As a result, their decisions tend to be more accurate, and collisions only occur in a limited number of cases, primarily when the hesitation or recklessness is excessive. For type 4, when TTC is greater than t_3 , the collision rates of the subjects approach 0. It suggests a minimal risk, enabling the subjects to safely cross the road with N decisions. This observed difference stems from the fact that the subjects do not rely on a single mode when making decisions. Instead, they dynamically adjust their behavior modes in response to the scenario urgency levels. To further identify the transition times (t_1 , t_2 , t_3 shown in Fig. 5) in pedestrian behavior modes, it is essential to examine both the spatiotemporal distribution of decisions and the variety of decision types.

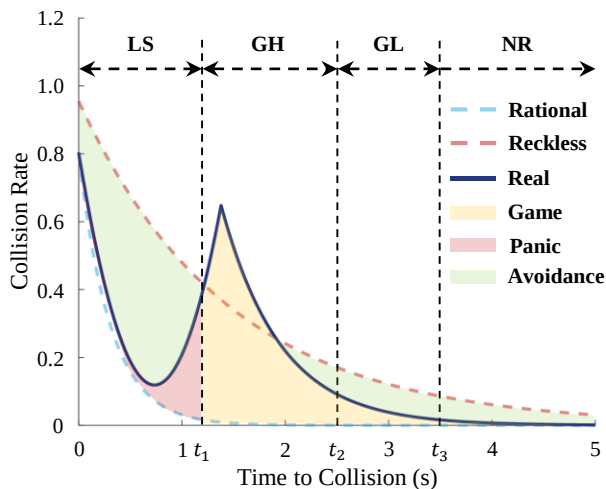


Fig. 5. Collision rate of pedestrians in different radical styles under different urgency levels (LS: Life-saving, GH: Game in high-risk, GL: Game in low-risk, NR: No-risk).

B. Spatiotemporal distribution of pedestrian decisions

1). Probability density of decisions

In each case, upon observing a hazardous vehicle, a significant change in a pedestrian's movement is considered as a decision. If a subject maintains N, it indicates his belief that crossing is safe and avoidance action is not necessary. The decision moment for the N was defined as the moment when the hazardous vehicle first enters the subject's visual field. We focused only on the subject's final decision (i.e., those made when his behaviors stabilized) because the final decision is closely linked to collision risk. As a result, a total of 1776 cases were analyzed to investigate the spatiotemporal distribution and types of subjects' decisions.

The probability density of decisions (Table BI) was

calculated to analyze the variability in pedestrian behavior modes under different levels of urgency. Considering that the cumulative probability of a pedestrian's final decision (i.e., whether to continue normal walking or avoid) during a single interaction is 100%, we analyzed the TTC intervals ranging from 0 to 4.5 s (Fig. 6a), with the purpose of identifying key transition times in pedestrian's decisions.

$$P_{TTC_i} = A_i \times \left(\frac{1}{\sigma_i \times \sqrt{2\pi}} \times e^{-\frac{(y-\mu_i)^2}{2 \times \sigma_i^2}} \right) \quad (3)$$

where P_{TTC_i} represents the probability density of decisions of pedestrians within the TTC_i interval, TTC_i denotes the time to collision intervals from 0 to 4.5 s, with $i \in \{1, 2, 3, 4, 5, 6\}$, A_i is the decision weight, representing the proportion of subjects who made their final decisions within the TTC_i , μ_i and σ_i are the mean and the standard deviation of the perpendicular distance between the subjects' final decision positions and the hazardous vehicles' centerline positions, y is the perpendicular distance.

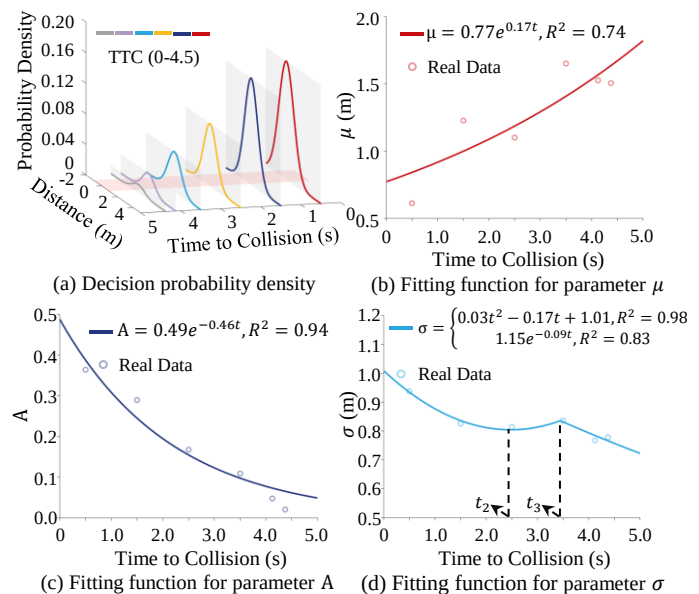


Fig. 6. Probability density of decisions. (a) Probability density of segmented decisions. (b) Fitting function for parameter μ . (c) Fitting function for parameter A . (d) Fitting function for parameter σ .

During interactions, a pedestrian can make decisions at any time and position. To represent this characteristic, we fitted parameters μ_i , A_i , and σ_i to ensure the spatiotemporal continuity of the probability density function of decisions (Fig. 6b, c, d). Thus, the probability density of a pedestrian making a final decision can be described as:

$$P(t, y) = \frac{A(t) \times \left(\frac{1}{\sigma(t) \times \sqrt{2\pi}} \times e^{-\frac{(y-\mu(t))^2}{2 \times \sigma(t)^2}} \right)}{\int_0^\infty A(t) dt} \quad (4)$$

where t represents the TTC, $P(t, y)$ is the probability density of final decisions in both time and space, $\mu(t)$ and $\sigma(t)$ are the functions of mean and standard deviation of the perpendicular distance between pedestrians' decision positions and the centerline positions of hazardous vehicles, $A(t)$ is the

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function that represents the weight of pedestrians' final decisions, and $\int_0^\infty A(t)dt$ is a normalization factor ensuring that the cumulative probability of final decisions equals to 100%.

Based on the above derivation, the probability that a subject's final decision and subsequent motion state remain unchanged is:

$$\text{Final}(t, y) = \int_y^\infty \int_t^\infty \frac{A(t) \times \left(\frac{1}{\sigma(t) \times \sqrt{2\pi}} \times e^{-\frac{(y-\mu(t))^2}{2 \times \sigma(t)^2}} \right)}{\int_0^\infty A(t)dt} dy dt \quad (5)$$

where t is the TTC and y is the pedestrian's perpendicular distance from the vehicle's centerline.

2). Analysis of parameters of probability density

The spatiotemporal positions where pedestrians make decisions in response to hazardous vehicles reveal their underlying logic, and the probability density parameters are described as:

$$\begin{aligned} \mu(t) &= 0.77e^{0.17t}, (R^2 = 0.74) \\ A(t) &= 0.49e^{-0.46t}, (R^2 = 0.94) \\ \sigma(t) &= \begin{cases} 0.033t^2 - 0.165t + 1.009, & t \leq 3.45, \\ & (R^2 = 0.98) \\ 1.15e^{-0.09t}, & t > 3.45, \\ & (R^2 = 0.83) \end{cases} \quad (6) \end{aligned}$$

The probability density of decisions (Fig. C.1) is characterized by three parameters, one of which is μ , representing the average perpendicular distance from the subjects' decision position to the centerline of the hazardous vehicle. As TTC increases, μ tends to increase from the vehicle's centerline. This shift suggests that, in less urgent scenarios, subjects can make their final decisions at a far distance from the vehicle, often resulting in N decisions. Conversely, in more urgent TTC scenarios, pedestrians will make avoidance decisions when the hazardous vehicle is closer. Parameter A represents the probability that the decision made by a pedestrian is indeed their final decision. It reveals that, in emergent scenarios, pedestrians have less time for adjustment, leading to a higher probability that his initial decisions will be final. The last parameter, σ , represents the standard deviation of a pedestrian's decision position relative to the vehicle's centerline. During the TTC from 0 to t_2 ($t_2 = 2.50$ s), σ exhibits a decreasing trend, suggesting that in high-urgency scenarios, the spatial dispersion of pedestrian decisions diminishes as TTC increases. It reflects that pedestrians shift from solely avoidance to balancing between N and avoidance decisions. In the cases when TTC is between t_2 and t_3 ($t_3 = 3.45$ s), σ increases as most pedestrians make N decisions when farther from hazardous vehicle, while a few in the vehicle's path adopt conservative avoidance behaviors. In the cases when TTC is greater than t_3 , the scenario presents minimal risk, allowing subjects to make N decisions from a greater perpendicular distance, which decreases σ . Therefore, t_2 and t_3 could serve as TTC boundaries between high and low risk and low and no risk.

C. Analysis of pedestrian decision types

1). Behavior modes under different urgent scenarios

The probability density of pedestrian decisions reveals the position in which pedestrians make decisions in the presence of a hazardous vehicle. Variations in the parameters, particularly the critical shifts indicated by σ , reflect the underlying logic of pedestrian decisions. Based on the key transition times, this section will analyze the modes of pedestrian behaviors under different urgent scenarios.

In addition to t_2 and t_3 , t_1 also marks a change in the subjects' collision rate, as it indicates the transitions from being lower than that of reckless pedestrians to being higher. A threshold-based classification method (Appendix D) was used to identify the t_1 value from 0 to 1.38 s (i.e., the time corresponding to the second turning point (Fig. 5) of subjects' collision rates). Therefore, the interaction between pedestrians and hazardous vehicles can be divided into four distinct types based on the three aforementioned turning times. It is hypothesized that pedestrian behavior modes vary when TTC and the urgency level are significantly different. We calculated statistical data for four specific decision types at each TTC interval to prove this hypothesis. Furthermore, we compared the decisions made by the subjects with those of the rational pedestrians to gain insights into the behavior mode at each TTC interval.

2). Identification and analysis of pedestrian behavior modes

Pedestrian decision types across different scenarios reflect their underlying decision logic. The threshold-based classification determined that setting $t_1 = 1.21$ s allows for the most accurate separation (91.9%) of avoidance decisions from normal walking decisions (Fig. D.1). As a result, the values of t_1 , t_2 , and t_3 are established, enabling the identification of the pedestrian behavior modes into four distinct types.

Life-saving mode from 0 to t_1 . In this mode, subjects mainly make avoidance decisions. The comparison in collision rates between rational pedestrians and subjects shows that the rational pedestrian's collision rate is significantly lower for N and S decisions than the subjects. In contrast, collision rates for F and B decisions are similar (Fig. 7a). This suggests that the optimal decisions in this mode are B or F. However, the subjects sometimes make sub-optimal N and S decisions. Within N instances (Fig. 7b, c), pedestrians involved in non-collision scenarios maintained a smaller distance from the vehicle's far side (perpendicular distance = -1 m) than those in collision scenarios (Median_{N_U} = -0.62 m, Median_{N_C} = 0.96 m). For the S behavior, non-collision cases generally stayed outside the vehicle's path, while collision cases pedestrian chose to stop inside the vehicle's path (Median_{S_U} = 1.95 m, Median_{S_C} = 0.40 m). The collision cases indicate that subjects made the N and S decisions due to the urgency of the scenarios, which led to panic or insufficient time to make alternative decisions and adjust their behaviors quickly. For

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the subjects demonstrating optimal decisions, those selecting B were positioned closer to the vehicle near-side (perpendicular distance = 1 m). In contrast, F were situated near the far-side ($\text{Median}_B = 0.91$ m, $\text{Median}_F = -0.59$ m). In the collision scenarios, the subjects were significantly closer to the vehicle's centerline ($\text{Median}_{B_C} = 0.30$, $\text{Median}_{B_U} = 1.19$ m, $\text{Median}_{F_C} = -0.07$, $\text{Median}_{F_U} = -0.77$ m), suggesting that the collisions in these scenarios were unavoidable. Consequently, HAVs should prioritize monitoring that area (Fig. 7d).

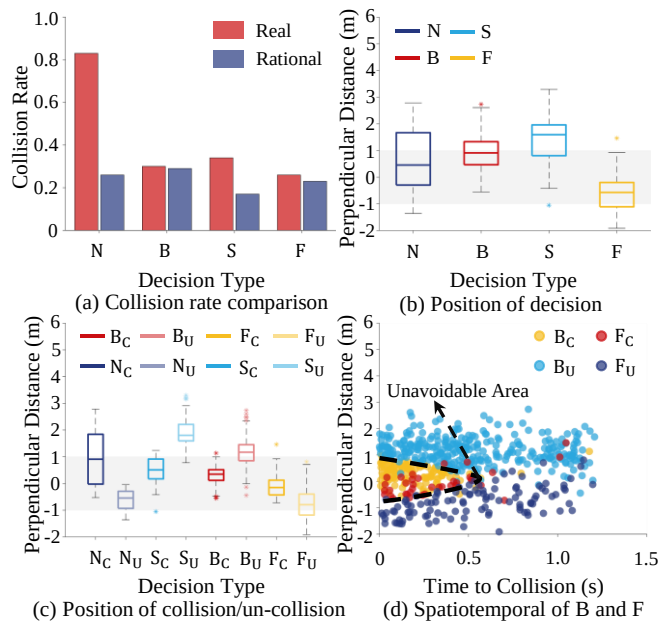


Fig. 7. Pedestrian decisions in the life-saving mode.

Game in high-risk mode from t_1 to t_2 . As TTC increases, pedestrians' desire to cross becomes more evident, as indicated by a higher proportion of subjects opting for N. Despite this, the scenarios remain high-risk, necessitating that subjects carefully balance their crossing desire and safety. In this mode, all collisions resulted from the N decision (Fig. 8a). Specifically, in B and S decisions, the subjects were closer to the vehicle near-side. The B decisions were primarily made in the vehicle's near-side ($\text{Median}_B = 0.97$ m), where the subjects perceived a threat and opted to retreat. The S decisions occurred in areas where the pedestrians had not yet reached the vehicle near-side ($\text{Median}_S = 2.31$ m). In cases of F, most subjects were already positioned within the vehicle's path ($\text{Median}_F = -0.09$). Although N could safely allow them to cross at these points (i.e., as a rational pedestrian would), the subjects opted for F, maintaining their motion direction while prioritizing safety (Fig. 8a, c). It suggests that some subjects, upon entering the vehicle's range, opted to F due to a perceived risk. As for N (Fig. 8b, d), the pedestrians' decision positions were usually outside of the vehicle's trajectory. However, in non-collision cases, they were already within the path ($\text{Median}_{N_U} = 0.90$ m). The subjects involved in collisions made the N decisions, choosing this risky decision to achieve higher scores ($\text{Median}_{N_C} = 1.65$ m).

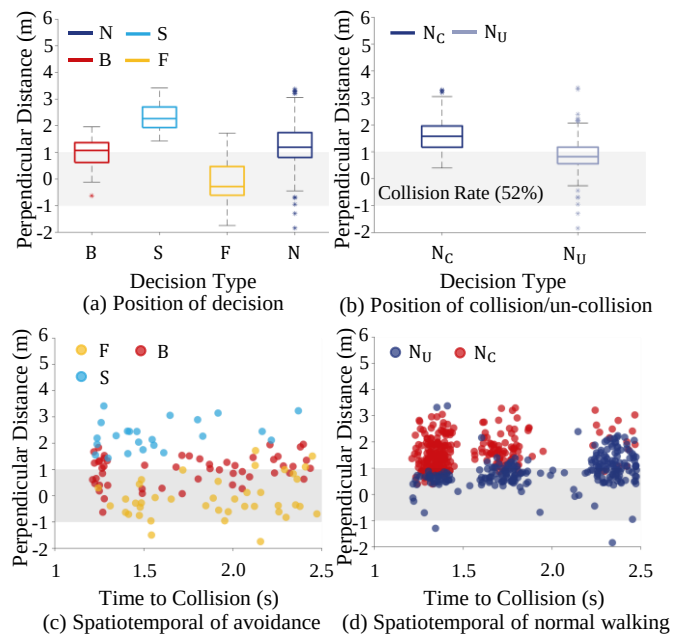


Fig. 8. Pedestrian decision in game in high-risk mode.

Game in low-risk mode from t_2 to t_3 . In low-risk scenarios, the collision rate for N decreased to 17% (Fig. 9b, d) and fewer subjects chose avoidance. Although most subjects could have safely crossed by N, some opted for a conservative decision ($\text{Median}_B = 1.14$ m; $\text{Median}_F = 0.10$ m, $\text{Median}_S = 2.37$ m), reflecting a strategy similar to that observed in high-risk scenarios (Fig. 9a, c). In contrast to high-risk scenarios, the subjects exhibiting the N decisions were positioned farther from the vehicle ($\text{Median}_{N_C} = 2.34$ m, $\text{Median}_{N_U} = 1.13$ m), irrespective of a collision, indicating that they had more time for crossing (Fig. 9b, d). It suggests that the subjects' decision basis gradually shifted from time urgent to perpendicular distance.

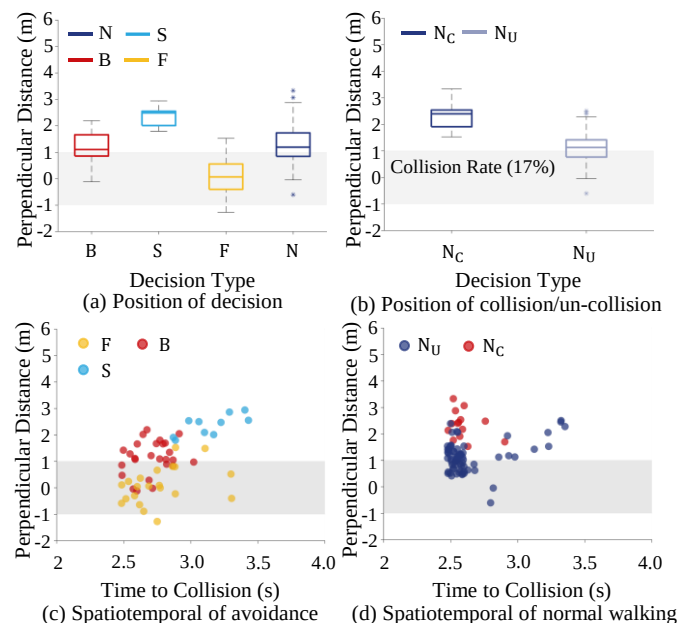


Fig. 9. Pedestrian decisions in game in low-risk mode.

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No-risk mode after t_3 . In no-risk scenarios, the collision rate tends to be 0. Most subjects walk normally, with only a few stopping (Median_N = 1.57 m, Median_S = 2.60 m). In these cases, when there is sufficient time, the subjects tend to cross safely at a greater perpendicular distance from the vehicle (Fig. 10a, b). Thus, HAVs may not need to prioritize protection during this period.

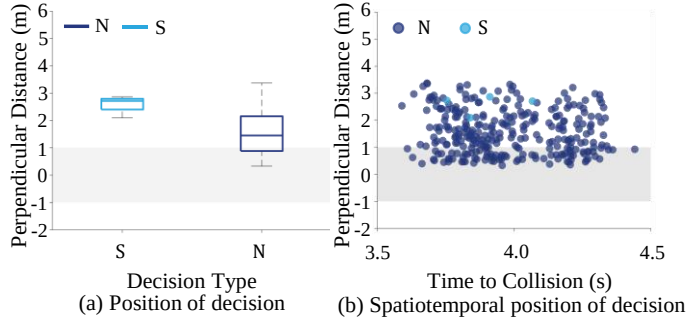


Fig. 10. Pedestrian decisions in no-risk mode.

IV. DECISION MODEL UNDER SAFETY-CRITICAL SCENARIO

A. Regression model for pedestrian decisions

The analysis of behavioral logic has revealed that pedestrians employ different behavior modes, which depends on the urgency of the scenarios. To effectively model these decisions, this section introduces a concept of expected safe distance and uses a regression model to describe how pedestrians choose decisions.

The expected safe distance is the perpendicular position that a pedestrian would reach if he/she continues to walk normally. Considering the current perpendicular distance to the vehicle and the TTC, the subject's expected safe distance can be described as:

$$e = y_t - 1.4t \quad (7)$$

where y_t represents the pedestrian's current perpendicular distance from the hazardous vehicle's centerline.

Occurrence of a collision can be determined by:

$$C = \begin{cases} \text{Collision}, & e \in (-1, 1) \\ \text{No collision}, & e \in (-\infty, -1) \cup (1, \infty) \end{cases} \quad (8)$$

We identified key factors that significantly influence pedestrian decisions through the principal component analysis. Utilizing these factors (horizontal distance between pedestrian and the vehicle (x_t), vehicle speed (v_t), and the expected safe distance (e_t)), a decision score function for pedestrians was defined as:

$$Z = \beta_0 X_t + \beta_1 V_t + \beta_2 E_t + c_0 \quad (9)$$

where Z denotes the decision score for pedestrian decision. X_t , V_t , and E_t are the standardized values of factors x_t , v_t , e_t , respectively. The weights of these factors are represented by β_0 , β_1 and β_2 , while c_0 signifies the bias term of the score function.

Based on (9), the score functions for different decisions are expressed as:

$$Z_{ij} = \tilde{X}_i^T \Theta_j + c_j \quad (10)$$

where i is the sample index, j represents the decision type. $\tilde{X}_i$ is the standardized parameter matrix, Θ_j is the weight matrix,

and c_j is the bias matrix.

And, the decision probability function for pedestrians is represented by:

$$p_{ij} = \frac{e^{Z_{ij}}}{\sum_{k=1}^n e^{Z_{ik}}} \quad (11)$$

where p_{ij} signifies the probability of pedestrian decisions, and n represents the total number of possible pedestrian decision types.

To estimate the parameters of the pedestrian decision score function, we employed a loss function:

$$\mathcal{L}_\omega = -\sum_{i=1}^N \omega_{A_i} \log(p_i, B_i) + \lambda \sum_{j=1}^n \sum_{l=1}^3 \Theta_{jl}^2 \quad (12)$$

where $\mathcal{L}_\omega$ represents the loss function and N denotes the total number of samples. The variable B_i indicates the actual decision of the samples. The weight of different decisions in the sample, represented by ω_{A_i} , is calculated as the normalized inverse frequency of each decision type. The $\log(p_i, B_i)$ refers to the logarithmic predicted probability of sample i for the B_i . Additionally, λ is the regularization coefficient, n signifies the number of pedestrian decision types, and l denotes the number of factors influencing pedestrian decisions.

Finally, the decision score equation parameters are estimated by (13), enabling the construction of the pedestrian decision model.

$$\min_{\theta, c} \mathcal{L}_\omega \\ \hat{B}_i = \arg \max_j p_{ij} \quad (13)$$

where θ and c are the matrices representing the weight and bias parameters to be estimated, while $\hat{B}_i$ indicates the pedestrian decision predicted by the model.

B. Performance of decision model

Pedestrian behavior modes vary with urgency level of scenarios. Consequently, a decision model necessitates estimating parameters from different urgent scenarios. Under the life-saving mode, only F and NF (i.e., B and behaviors where pedestrians stop outside vehicle's path) are predictable. The model's parameters and performance are detailed in Table EI and Fig. 11a. The model shows a higher precision for NF (97%) compared to 75% precision for F. This difference in precision between NF and F can be attributed to the tendency of pedestrians in the unavoidable collision area to randomly choose B and F decisions. Furthermore, pedestrians show a prominent preference for NF, with 78% opting for NF over 22% for F. These inconsistent proportions lead the model to misclassify B as F (Fig. 11a). During the game in high-risk mode, human subjects exhibited four distinct decision behaviors. Notably, the model achieves a prediction precision of 98% for N, a rate significantly higher than the precision for avoidance decisions (80%). This is because, in this mode, 81% of pedestrians tend to choose risky N decisions, while only 19% opt for avoidance behaviors. Therefore, the model misclassified some N as avoidance (Fig. 11b). As for the prediction of avoidance decisions, the model exhibits balanced prediction precision (83%) for S and B. However, the precision for F is comparatively lower (74%). This can be attributed to the inherent randomness in pedestrian decisions, where they may alternate between B and F within hazardous

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vehicle's path. In some situations, this characteristic leads the model to misclassify some B behaviors as F (Fig. 11d). In the game in low-risk mode, the model shows a higher prediction precision for N (96%) than avoidance (76%). It can be attributed to the increased randomness in pedestrian decisions during this phase, where the lower risk allows for more variability in choosing between N and avoidance behaviors. It causes miss prediction of the model, where some N are incorrectly identified as avoidance behaviors (Fig. 11c). The model achieves significantly higher precision (88%) and recall (90%) for S, compared to B (78%, 69%) and F (80%, 77%). This can be attributed to the fact that pedestrians who opt to stop typically do so at a position significantly farther from the hazardous vehicle's near-side. In contrast, within the central range of vehicle's path, pedestrians can choose either F or B to avoid collisions successfully. This flexibility results in greater randomness between B and F. Consequently, the model mispredicts B and F as one another (Fig. 11e). Regarding the no-risk behavior mode, most pedestrians opt for N. Only a few conservative pedestrians choose to stop, and these S behaviors are inherently unpredictable.

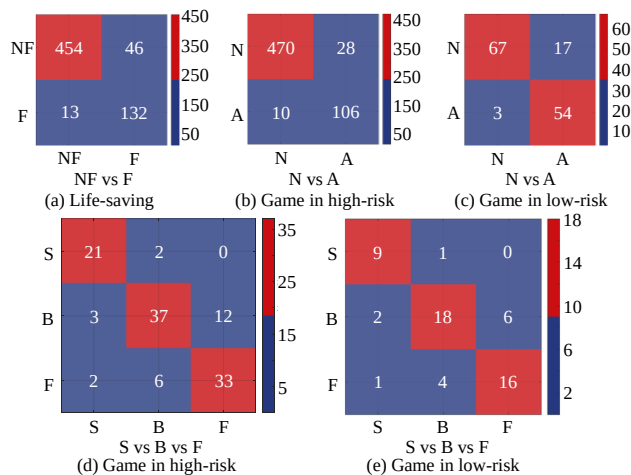


Fig. 11. Confusion matrix of pedestrian decision model under different urgent scenarios (NF: No Forward, A: Avoidance).

To evaluate the effectiveness of the decision model, we conducted a comparative analysis with widely used machine learning models, including support vector machine (SVM) and decision tree (DT) models (Table EI). The results have revealed that, in binary prediction tasks, both SVM and our model surpasses the performance of the DT model. Furthermore, in three-class prediction tasks, our model demonstrates superior performance. It suggests that our model, which is grounded in behavior modes, can effectively capture pedestrian decision to some extent.

V. DISCUSSION

Using a virtual reality experiment, we investigated and modeled pedestrian decisions in safety-critical traffic scenarios. The findings of this study provide insights into pedestrians' responses when encountering hazardous vehicles, enriching our understanding to the interactions between pedestrians and hazardous vehicles. Such understanding establishes a foundation for developing future pedestrian safety algorithms, and may enable future HAVs to interact

with pedestrians in a manner closely resembling human behaviors.

A. Significance and possible applications of the experiment

This research focuses on pedestrian's decisions upon noticing a hazardous vehicle. We emphasized the underlying mechanisms that drive pedestrian decisions, while previous studies of our lab analyzed pedestrian's motion ability and kinematic postures [16], [36]. Due to the high collision risk in the focused scenarios, a VR-based pedestrian simulator was implemented to ensure the subjects' safety. The high fidelity of the virtual experimental setup, enhanced by the scenario parameters based on real accident statistics [11], [13], ensures the experiment's realism and validity. In the experiment, the structured scenario design resulted in consistent data quality, aiding the analysis of pedestrian decisions. Additionally, the virtual collisions posed less perceived threat to the subjects than real-world scenarios. To address this issue, we implemented a reward-punishment mechanism to simulate the real threat. Moreover, the collision rates associated with each decision type were within reasonable limits, demonstrating the effectiveness of the approach used in this study.

We constructed a human-annotated dataset focused on safety-critical scenarios. The information is typically scarce and often incomplete in the real world. The dataset includes pedestrian and vehicle positions, speed, acceleration, collision information, and record files, which were commonly featured in the existing researches [2]. We also recorded other information through the detailed manual annotations of pedestrian decision data, including the critical moments such as when a pedestrian first perceived a hazardous vehicle, the timing of changes in his movement state, and the corresponding decisions. These data enable an in-depth analysis of pedestrian decision mechanisms and support the development of the decision models. Furthermore, we collected pedestrian visual data in the experiment, such as the gaze points that could aid in constructing visual attention models. Analyzing scenario information and processing from a visual attention perspective can facilitate the development of high-level pedestrian models [10]. Finally, each subject reviewed first-person video replays of his interactions with the hazardous vehicles and provided explanations of his key decision. When technologies becoming available in the future, these verbal accounts may be integrated with large language models to help develop interpretable pedestrian models, enhancing HAVs' ability to understand and interact more effectively with pedestrians.

B. Cognitive insights into pedestrian behavior modes

Three types of pedestrians are identified: reckless pedestrians, real pedestrians (subjects), and rational pedestrians. Reckless pedestrians are referred to those whose alignments with the assumptions commonly found in safety algorithms. Those algorithms presume that pedestrians maintain a constant speed, which is generally reasonable as reckless pedestrians are considered to have the highest collision rates. However, real pedestrians often experience higher collision rates during the game in high-risk TTC interval than reckless pedestrians. This difference arises from

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the mistakes or hesitation of real pedestrians. Unfortunately, the existing pedestrian protection algorithms do not adequately account for this phenomenon, leading to an underestimation of pedestrian risks during this critical period. Meanwhile, real pedestrians demonstrate significantly lower collision rates during the life-saving, game in low-risk, and no-risk TTC interval than reckless pedestrians. This is because they can assess risks and make avoidance decisions. Consequently, it may be inappropriate for self-driving vehicles to maintain the current pedestrian protection strategies during the game in low-risk and no-risk TTC interval. Such strategies could cause unnecessary braking by HAVs, potentially causing rear-end collisions and significantly reducing traffic efficiency.

The decisions of pedestrians in safety-critical scenarios are complex and crucial for safety. This study has identified four types of pedestrian behavior modes based on their decision types, spatiotemporal probability density distributions, and collision rates. In the no-risk mode, pedestrians perceive no threat from oncoming vehicles, which allows them to focus exclusively on the rewards and choose to N. As pedestrians are in low-risk mode, pedestrians begin to balance their crossing desire with safety considerations. In this mode, the low time pressure enables pedestrians to gather ample scenario cues. As a result, they can make near-optimal decisions while accepting low risk. These decisions align with the underlying logic proposed by existing study [25]. When pedestrians engage in high-risk mode, they experience increased time pressure, necessitating rapid decisions. The urgency often results in limited collection of scenario cues [26], thereby increasing the likelihood of radical decisions and, subsequently, higher collision rates [37], [38]. In the experiments of this study, such pressure led pedestrians to choose N, while N carried a heightened risk of collision penalties. Finally, in the life-saving mode, we were surprised to see that the collision rates of real pedestrians began to approximate those of rational pedestrians. This shift can be attributed to the increased time pressure, which causes pedestrians to reach their cognitive load. They abandon the complex four-choice mode and transition to a simpler three-choice mode. This phenomenon is consistent with the findings, which concluded that under sufficiently high time pressure, humans deviate from the traditional trade-off and execute a quicker and more effective strategy [39]. To our knowledge, our study is the first study to use cognitive theories to explain pedestrian behaviors in safety-critical scenarios.

C. Engineering applications of the pedestrian decision model

For driver-assistance tasks, several pedestrian models have been proposed in the literature, such as detection and intent recognition [2]. These models have proven effective in facilitating interactions between human drivers and pedestrians. Along with the prevalence of HAVs, the growing need of safe interaction between HAVs and pedestrians shift necessitates the development of models that can more accurately capture and predict pedestrian decisions. Based on the expected safety distance and using a linear regression approach, in this study, we have proposed an initial pedestrian decision model. This model aims to reflect real pedestrian

decisions, thereby enabling HAVs to develop more adaptive safety algorithms. For instance, in the interactions between HAVs and pedestrians, different risk types require distinct safety strategies. HAVs may engage with pedestrians in case no-risk or low-risk types are identified. The situations require minimal interventions, such as no braking or light braking. In case of high-risk types, HAVs must implement more conservative safety measures. In the life-saving type, the focus shifts to pedestrian avoidance decisions, requiring vehicle movement adjustments, such as full braking and steering away from the pedestrians. Additionally, in scenarios where pedestrians do not react or stop within the vehicle's path, the algorithm should pre-activate pop-up hood to mitigate the injury severity in case of collision [40].

Another potential application is enhancing physical and virtual testing for HAVs [41]. Presently, pedestrian dummies used in physical safety tests are programmed to move at constant speeds under specific conditions. This assumption, depicted in Fig. 5, mirrors reckless pedestrian decisions, resulting in significant discrepancies between the test results and actual interactions. By implementing a more realistic pedestrian model, these discrepancies can be minimized, thereby improving the accuracy and reliability of HAV safety assessments. More importantly, integrating pedestrian models with virtual testing is crucial as it facilitates large-scale, multi-scenario testing in simulated environments [42]. Virtual testing significantly reduces costs while enhancing the realism of HAVs testing.

D. Limitations and future applications

Several limitations of the present study are discussed as follow. First, the experiment exclusively recruited young male subjects. Other demographic groups, such as child, elderly and female, whose behaviors may vary due to differences in physical ability and cognitive styles, were not included in the study. Second, while the study focused on two common safety-critical scenarios, it is important to examine other high-risk locations, such as parking lots and bus stops. Additionally, the study only investigated the effects of spatiotemporal pressure from hazardous vehicles on pedestrian decisions, neglecting the potential influence of vehicle braking or steering actions. Moreover, pedestrian decisions in interactions with hazardous vehicles are inherently continuous and dynamic, yet this study did not model such process. Future research should broaden the range of subjects and scenarios to address these limitations and develop a more comprehensive pedestrian model. That would assist HAVs in understanding pedestrian decisions under safety-critical scenarios and support the development and testing for safety algorithms.

VI. CONCLUSIONS

This study identified and modeled pedestrian decisions during interactions between pedestrians and vehicles in safety-critical scenarios. We designed and conducted a virtual experiment with pedestrian-in-loop to address the challenge of gathering pedestrian behavior data under safety-critical scenarios. We analyzed pedestrian behavior modes under

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varying time pressures and created an annotated dataset. The study has revealed a significant trend: as urgency increases, reflected by how late a pedestrian sensing an oncoming hazardous vehicle, pedestrian behavior modes will shift from prioritizing road crossing to personal safety. There exist four distinct behavior modes: no-risk, game in low-risk, game in high-risk, and life-saving. By characterizing these modes, we proposed a preliminary pedestrian decision model. The model achieved an average F1-scores of 0.88 and 0.78 for predicting whether and how pedestrian avoid a collision. The pedestrian behavior modes we identified from the experiment and the pedestrian decision model we built in this study have laid a groundwork for developing and testing pedestrian safety systems, thereby aiding HAVs to better understand pedestrian behaviors and mitigate injury risks in urban traffic environments.

APPENDIX

Appendix A. Information of the virtual reality experiment

TABLE AI
VARIABLE OF THE EXPERIMENT

Experiment Variable				
Scenario Type	Mid-Block		Crosswalks	
Obstacle Vehicle	Yes		No	
Vehicle Direction	Left		Right	
Vehicle Velocity (km/h)	15	25	35	45
Perpendicular distance (m)	0-1.5		1.5-3	
Horizontal distance (m)	18			

Appendix B. Parameters of decision probability density

TABLE BI
PARAMETERS OF DECISION PROBABILITY DENSITY UNDER DIFFERENT URGENCY LEVELS

Parameter	TTC_1	TTC_2	TTC_3	TTC_4	TTC_5	TTC_6
A_i	0.36	0.29	0.17	0.11	0.05	0.02
μ_i	0.61	1.23	1.10	1.65	1.53	1.51
σ_i	0.94	0.83	0.81	0.84	0.77	0.78
P-value	0.31	0.09	0.16	0.38	0.25	0.21

$TTC_1(0-1)$, $TTC_2(1-2)$, $TTC_3(2-3)$, $TTC_4(3-4)$, $TTC_5(4-4.25)$, $TTC_6(4.25-4.5)$.

Appendix C. Pedestrian decision probability density

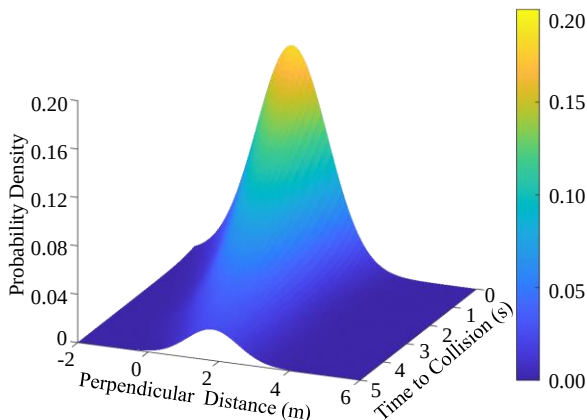


Fig. C.1. Pedestrian decision probability density.

Appendix D. Threshold-based classification method

A threshold-based classification method was used to identify the t_1 value from 0 to 1.38 s (i.e., the time corresponding to the second turning point (Fig. 5) of subjects' collision rates).

$$d = \begin{cases} 1, & \text{if } 0 \leq t \leq t_1 \\ 0, & \text{if } t_1 < t \leq 1.38 \end{cases} \quad (D1)$$

where d represents the classification rule.

$$R(t) = \frac{1}{N} \sum_{i=1}^N I(d_i = \hat{d}(t_i, t)) \quad (D2)$$

$$I(d_i = \hat{d}(t_i, t)) = \begin{cases} 1, & \text{if } d_i = \hat{d}(t_i, t) \\ 0, & \text{if } d_i \neq \hat{d}(t_i, t) \end{cases}$$

where $R(t)$ is the accuracy of the classification under the threshold of t , N is the number of pedestrian decisions within 1.38 s, d_i is the subjects' actual decision: 1 for avoidance and 0 for N, t_i is the subjects' decision time, I is the indicator function.

$$t^* = \arg \max_t R(t) \quad (D3)$$

where t^* represents the TTC threshold for maximum accuracy and corresponds to the t_1 .

Therefore, we determined an optimal time threshold within the TTC range of 0-1.38 s to distinguish between N and avoidance decisions. The result shows that setting $t_1 = 1.21$ s yields a maximum classification accuracy of 91.9%, as illustrated in Fig. D1.

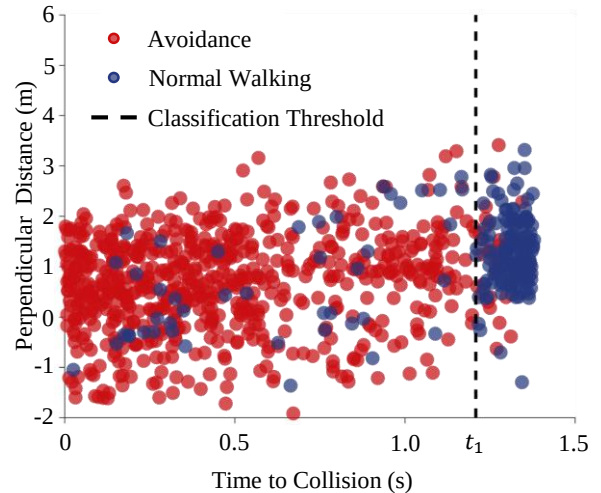


Fig. D.1. Optimal time threshold between normal walking and avoidance decisions.

Appendix E. Decision model parameters and performance

Different behavior modes correspond to different decision types. Therefore, it is necessary to separately estimate the model parameters and compare the performance between our model with common machine learning models (Table EI).

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TABLE EI
PARAMETERS AND PERFORMANCE OF DECISION MODEL

Decision	Decision	β_0	β_1	β_2	c_0	Precision	Recall	F1	Ave F1
LS_{our}	NF	0.19	-0.44	2.22	1.02	0.97	0.91	0.94	0.88
	F	-0.19	0.44	-2.21	-1.03	0.75	0.91	0.82	
LS_{SVM}	NF					0.95	0.96	0.95	0.90
	F					0.86	0.81	0.84	
LS_{DT}	NF					0.94	0.94	0.94	0.86
	F					0.79	0.79	0.79	
GH_{our}	N	1.12	0.67	-0.64	0.63	0.98	0.94	0.96	0.90
	A	-1.12	-0.67	0.64	-0.63	0.80	0.91	0.85	
GH_{SVM}	N					0.96	0.96	0.96	0.89
	A					0.82	0.83	0.82	
GH_{DT}	N					0.95	0.93	0.94	0.84
	A					0.73	0.77	0.75	
GH_{our}	S	0.60	-0.27	2.95	-1.39	0.83	0.92	0.85	0.79
	B	-0.58	-0.11	-0.10	1.06	0.83	0.71	0.76	
GH_{SVM}	F	-0.02	0.38	-2.85	0.36	0.74	0.81	0.77	0.76
	S					0.89	0.66	0.71	
GH_{DT}	B					0.74	0.83	0.78	0.74
	F					0.81	0.78	0.80	
GL_{our}	N	0.33	1.10	0.04	0.55	0.96	0.80	0.87	0.86
	A	-0.33	-1.10	-0.04	-0.56	0.76	0.95	0.84	
GL_{SVM}	N					0.99	0.80	0.88	0.87
	A					0.77	0.98	0.86	
GL_{DT}	N					0.84	0.87	0.85	0.80
	A					0.81	0.73	0.76	
GL_{our}	S	1.45	-0.56	1.55	-1.25	0.88	0.90	0.85	0.77
	B	-1.05	0.57	0.30	0.76	0.78	0.69	0.72	
GL_{SVM}	F	-0.40	-0.01	-1.85	0.49	0.80	0.77	0.74	0.72
	S					0.90	0.70	0.73	
GL_{DT}	B					0.72	0.69	0.69	0.61
	F					0.73	0.77	0.72	
	S					0.60	0.40	0.47	
	B					0.66	0.66	0.63	
	F					0.73	0.77	0.72	

LS: Life-saving, GH: Game in high-risk, GL: Game in low-risk; NF (No Forward), F (Forward Accelerating), N (Normal Walking), S (Stop), B (Backward Avoidance).

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